0710 BIOLOGY 1

GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2024 **ADVANCED LEVEL**

Centre Number	
Centre Name	
Candidate Number	
Candidate Name	

Mobile phones are NOT allowed in the examination room.

MULTIPLE CHOICE QUESTION PAPER

Duration: One and a Half Hours

INSTRUCTIONS TO CANDIDATES

Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.

1. USE A SOFT HB PENCIL THROUGHOUT THE EXAMINATION.
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the examination begins:

3. Check that this question booklet is headed "**Advanced Level – 0710 Biology 1**".
4. Fill in the information required in the spaces above.
5. Fill in the information required in the spaces provided on the answer sheet using your HB pencil:
   Candidate Name, Exam Session, Subject Code and Candidate Identification Number.
   Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.

How to answer the questions in this examination

6. Answer **ALL** the **50** questions in this Examination. All questions carry equal marks.
7. Each question has FOUR suggested answers: **A, B, C** and **D**. Decide on the best answer that is correct. Find the number of the question on the Answer Sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen.
   For example, if **C** is your correct answer, mark **C** as shown below:
   [A] [B] [~~C~~] [D]
8. Mark only one answer for each question. If you mark more than one answer, you will score a zero for that question If you change your mind about an answer, erase the first mark carefully, then mark your new answer.
9. Avoid spending too much time on any one question. If you find that a question is difficult, move on to the next question. You can come back to this question later on.
10. Do all rough work in this booklet, using, where necessary, the blank spaces in the question booklet.
11. **At the end of the examination, the invigilator shall collect the answer sheet first and then the question booklet after. DO NOT ATTEMPT TO LEAVE THE EXAMINATION HALL WITH IT.**

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1. C_4 and CAM plants store CO_2 in the form of:
   A Pyruvate.
   B Glycerate phosphate.
   C Rubulose biphosphate.
   D Malate.

2. Which of the following refers to a temporary, reversible response to a non-directional stimulus?
   A Phototropism.
   B Nastic movement.
   C Thigmotropism.
   D Geotropism.

3. In what direction does an electrical signal move through a neurone?
   A Dendrite, cell body, then axon.
   B Axon, dendrite, then cell body.
   C Cell body, dendrite, then axon.
   D Axon, cell body, then dendrite.

4. The animal that has a four-chambered heart is:
   A Snake.
   B Turtles.
   C Dolphins.
   D Sharks.

5. The diploid number for an organism is six (6). One homologous pair underwent non-disjunction during meiosis I, how many chromosomes would be present in the daughter cells at the end of meiosis I?
   A Both daughters would have three.
   B One would have two and the other three.
   C One would have one and the other five.
   D One would have two and the other four.

6. Which of the following hormones would stimulate the testes to produce androgens?
   A Testosterone.
   B Luteinizing hormone.
   C Follicle stimulating hormone.
   D Gonadotrophin releasing hormone.

7. The water potential in a cell in which the solute potential is – 0.38kPa and the pressure potential is 0.25kPa is:
   A -0.63.
   B -0.13.
   C 0.13.
   D 0.63.

8. In a developing seed, the suspensor:
   A is formed from the cotyledons.
   B transfers nutrients from the parent plant to the embryo.
   C forms a protective coat.
   D develops into root and shoot meristems.

9. Homologous chromosomes separate during:
   A Anaphase I.
   B Anaphase II.
   C Telophase I.
   D Prophase II.

10. The sliding filament mechanism:
   A causes thick and thin filaments to slide towards the center of the H-band, bringing the Z-lines closer together.
   B causes thick and thin filaments to slide towards the center of the I-band, bringing the M-lines closer together.
   C causes thick and thin filaments to slide towards the center of the M-lines bringing the Z-lines closer together.
   D causes thick and thin filaments to slide towards the center of the A-band, bringing the Z-lines closer together.

11. Which of the following activities would cause the stomata to close?
   A Low concentrations of CO_2 in leaf air space.
   B Increased daytime light levels.
   C Abscisic acid production in leaves.
   D Absorption of moisture in the soil.

12. If a stimulus resulted in amplification of the response, this would be an example of:
   A Negative feedback.
   B Positive feedback.
   C Homeostasis.
   D Stimulation.

13. Nerve impulses:
   A sometimes travel in both directions along an axon.
   B always travel with the same intensity along the length of the axon.
   C diminish in intensity as they travel along the axon.
   D are always produced, no matter the intensity of the stimulus.

14. If one temporal lobe in the head is damaged, you would expect:
   A blindness in one eye.
   B total blindness.
   C loss of hearing in one ear.
   D partial loss of hearing in both ears.

15. Pituitary dwarfism is caused by:
   A hypersecretion of cortisol.
   B hyposecretion of somatotrophin.
   C hyposecretion of thyroid hormone.
   D hyposecretion of insulin-like growth factor.

16. Which of the following is an inorganic enzyme helper?
A Enzyme activator.
B Coenzyme.
C Cofactor.
D Prosthetic group.

17. A primary difference between mitosis in eukaryotic cells and binary fission in prokaryotes is that:
A binary fission lacks cytokinesis.
B binary fission is not regulated.
C binary fission does not use centrioles.
D DNA replication occurs only in mitosis.

18. How would you describe the ecological status of a forest that has been left undisturbed for 100 years?
A Pioneer community.
B Primary successional community.
C Plagioclimax community.
D Climax community.

19. Ashu has type A blood, and Bih has type B blood. What are the possible blood types of their children if Ashu and Bih get married?
A Type A and B.
B Type A, B, AB and O.
C Type A, B, and AB.
D Type AB.

20. A fragment of DNA has 60 Guanine bases and 30 Adenine bases. What will be the total number of deoxyribose sugar molecules in this fragment?
A 30.
B 60.
C 90.
D 180.

For questions 21-28, one or more of the responses is/are correct. Choose:
A. If (i, ii, and iii) are correct.
B. If (i and iii) are correct.
C. If (ii and iv) are correct.
D. If only (iv) is correct.

21. Which of the following takes place as red blood cells circulate into the region of actively metabolising tissues such as a muscle?
i Increase binding of carbon dioxide to haemoglobin.
ii Increase in release of oxygen from haemoglobin.
iii Increase in bicarbonate ion concentration in plants.
iv Movement of chloride out of red blood cells.

22. Many cellular poisons including cyanide are toxic to cells because:
i they bind reversibly to enzyme active site.
ii they are competitive inhibitors.
iii they are allosteric inhibitors of essential enzymes.
iv they bind on essential enzymes irreversibly.

23. Yogurt is a bacterial culture. As the culture gets older, it becomes increasingly sour due to the accumulation of lactic acid. Which of these stage(s) is (are) responsible for producing this acid?
i Krebs's cycle.
ii Glycolysis.
iii Electron transport system.
iv Fermentation.

24. In a closed blood circulatory system:
i the blood is confined to blood vessels.
ii the blood directly baths the organs.
iii substances are exchanged between the blood and the interstitial fluid.
iv the heart releases haemolymph into tissues.

25. For the cross AA x aa, if the F_1 generation was a different phenotype than either of the parental individuals, which of the following is the best and simplest explanation?
i Multiple alleles.
ii Incomplete dominance.
iii Linkage.
iv Codominance.

26. Body temperature in animals can be increased by:
i muscle contractions.
ii drinking of alcohol.
iii puffing up feathers.
iv increasing blood flow of the ears.

27. Some strains of the common bacterium *Staphylococcus aureus*:
i are resistant to antibiotic such as Penicillin.
ii produce an enzyme penicillinase which deactivates penicillin.
iii undergo mutation to produce individual bacterium.
iv are used in the production of bread.

28. Spores produced by plants:
i grow into gametophytes.
ii are haploid.
iii are produced by meiosis.
iv grow into the sporophyte plant.

29. Which of the following group of organisms is a population?
A All chicks in poultry.
B All plants in the school garden.
C All birds in the forest.
D All insects living on a large tree.

30. The following can be considered a tertiary structure of protein.
A A polypeptide chain folded and held by hydrogen bonds.
B The specific sequence of amino acids in a polypeptide.
C The pleated sheet of a polypeptide.
D A polypeptide chain folded and held in place by ionic bonds and hydrophobic interactions.

31. Phenotypic similarities that evolved independently in different lineages are called:
A Homologous characters.
B Evolutionary mosaic.
C Analogous characters.
D Morphological characters.

32. An observation of Darwin, to explain his theory of natural selection is that:
A certain hereditary characteristics allow some individuals to survive longer and reproduce more than others.
B most variations in an individual have a genetic basis and so are inherited by subsequent generations.
C a population's characteristics will change over time.
D individuals within a population compete for limited resources.

33. Complementary DNA (cDNA) is made by copying:
A DNA.
B mRNA.
C rRNA
D Plasmids.

34. This pigment gives brown algae their colour is:
A Phycobilins.
B Chlorophylls.
C Fucoxanthin.
D Carotenoids.

35. Choose the DNA strand which is synthesized in a discontinuous manner.
A Leading strand.
B Discontinuous strand.
C Lagging strand.
D Okazaki fragment strands.

36. In isometric growth:
A organs of the body grow at the same rate.
B increase in size is accompanied by change in shape.
C growth is intermittent.
D organs of the body do not grow at the same rate.

37. In which habitat do animals that secrete uric acid crystals live?
A Freshwater.
B Marine.
C Terrestrial and very dry.
D Terrestrial and very wet.

38. In a response to inflammation, which leukocytes are killed along with the pathogen forming pus at the infection site?
A Leucocytes.
B Basophils.
C Eosinophils.
D Neutrophils.

39. Which of these biomes is characterised by a hot humid weather with no seasonal changes?
A Tundra.
B Taiga.
C Deciduous forest.
D Tropical rainforest.

40. In genetic engineering, which vector is most suitable for carrying large pieces of DNA?
A Bacteriophage.
B Bacterium.
C Plasmid.
D Recombinant DNA.

41. Amongst which organisms is the competition for light and nutrients most severe?
A Closely related organisms growing in the same habitat.
B Closely related organisms growing in the same niche.
C Distantly related organisms growing in the same niche.
D Closely related organisms growing in different niches.

For questions 42 to 50 there are two statements. Read through the statements and then choose:

A if both statements are true and the second explains the first.
B if both statements are true but the second does not explain the first
C if the first statement is true and the second is false
D if the first statement is false and the second is true.

	First Statement	Second Statement
42.	Ethylene is a growth hormone in plants.	Ethylene promotes senescence, abscission and fruit ripening.
43.	Glucose is the first product of the Calvin cycle.	The Calvin cycle produces the building block for all other cellular organic molecules.
44.	Mitochondria and chloroplast contain ribosomes that resemble those of all prokaryotes.	Not all eukaryotic ribosomes are found on the surface of rough endoplasmic reticulum.
45.	The ideal soil pH effective for most plants to grow is between 6.0 to 7.0.	Soil pH does not affect the absorption of mineral salts by plants.
46.	Photorespiration is a wasteful process.	C_3 plants carry out photorespiration, though no ATP is formed during the process.
47.	Fresh water fish produce ammonia as nitrogenous waste.	High concentrations of ammonia in fresh water, makes it difficult for fish to eliminate ammonia from their bodies.
48.	Cholecystokinin is an enzyme secreted by the intestinal walls.	Cholecystokinin stimulates the pancreatic cells to secrete digestive juices and induces the gall bladder to release bile.
49.	All plants flower, at high concentrations of the phytochrome P_{FR}.	P_{FR} is the physiologically active form of the phytochrome.
50.	Sickle cell anaemia is an example of chromosome mutation.	Chromosome mutation results in a change of the number or structure of the chromosome, which may be lethal to the organism.

STOP

GO BACK AND CHECK YOUR WORK